

Project_1

SCENARIO :

Hospital Management System

The ERD diagram represents a hospital management system that organizes the relationships between patients, nurses, doctors, departments, rooms, appointments, bills, and medicines.

Patient Entity The central entity in the system is the Patient, which includes attributes such as Patient_ID, name (first and last), date of birth, gender, address, phone number, and age (which is derived). Patients are connected to multiple entities through different relationships.

Rooms & Nursing Care A patient can stay in a room through a "Stay in" relationship, where each room has attributes like Room_ID, room number, and type. Nurses are responsible for caring for patients through a many-to-many relationship called "Care for," meaning one nurse can care for multiple patients and each patient can be cared for by multiple nurses. Nurses can also supervise rooms through a relationship called "Supervises", where one nurse can supervise multiple rooms and each room is supervised by one nurse. Nurses are also assigned to departments, and each department has attributes such as Department_ID, name, phone, and location.

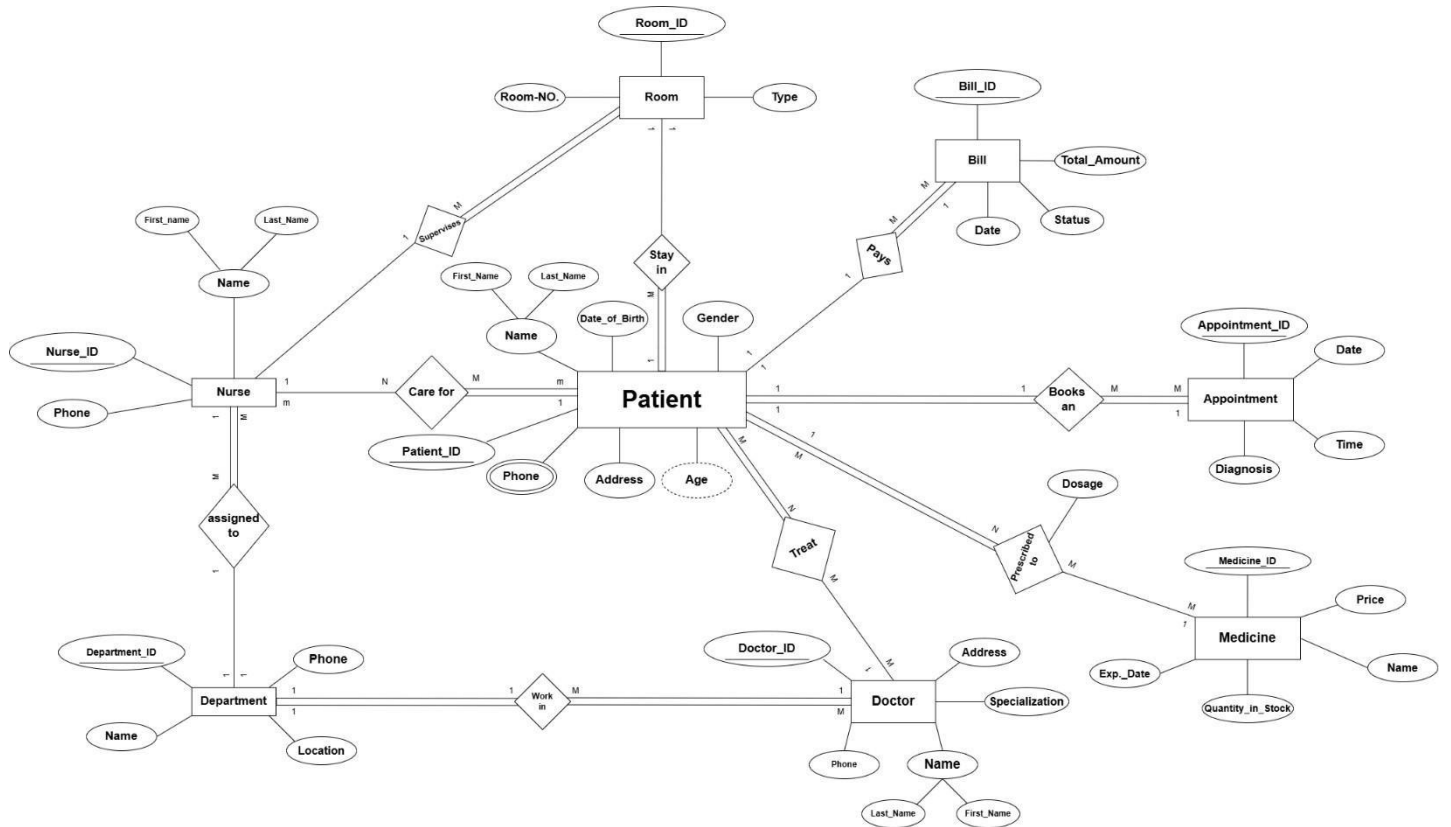
Doctors & Medicines Doctors are another important entity, identified by Doctor_ID and attributes like name (first and last), phone, address, and specialization. Doctors work in departments and treat patients through a "Treat" relationship. Additionally, doctors prescribe medicines to patients through a many-to-many relationship "Prescribe," where medicines have attributes such as Medicine_ID, name, price, dosage, expiration date, and quantity in stock.

Appointments & Bills Patients can also book appointments, where each appointment has an Appointment_ID, date, time, and diagnosis. Furthermore, patients are linked to bills through a "Pays" relationship, where each bill includes Bill_ID, total amount, date, and status.

Conclusion Overall, this ERD clearly models the hospital system by showing entities, attributes, and relationships, ensuring efficient data organization and management.

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ERD :



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Schema :

